
DOVA: Deliberation-First Multi-Agent Orchestration for Autonomous Research Automation

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Abstract

Large language model (LLM) agents have demonstrated remarkable capabilities in tool use, reasoning, and code generation, yet single-agent systems exhibit fundamental limitations when confronted with complex research tasks demanding multi-source synthesis, adversarial verification, and personalized delivery. We present DOVA (**Deep Orchestrated Versatile Agent**), a multi-agent platform introducing three innovations: (1) *deliberation-first orchestration*, where explicit meta-reasoning precedes tool invocation, informed by a persistent user model and entity-aware conversation context; (2) *hybrid collaborative reasoning*, a composable three-phase pipeline unifying ensemble diversity, blackboard transparency, and iterative refinement; and (3) *adaptive multi-tiered thinking*, a six-level token-budget allocation scheme reducing inference cost by 40–60% on simple tasks while preserving deep reasoning capacity. We formalize the core algorithms, present an architectural ablation study across seven system configurations, and analyze the contribution of each component to answer confidence, source coverage, and token efficiency.

1 Introduction

The rapid advancement of large language models (LLMs) [Brown et al., 2020, Anthropic, 2024a] has enabled a new generation of autonomous agents capable of reasoning, tool use, and multi-step planning [Yao et al., 2023b, Schick et al., 2023]. However, deploying these agents for *complex research automation*—where a single query may require searching academic databases, analyzing code repositories, cross-referencing model registries, and synthesizing findings with citations—exposes several limitations of single-agent architectures:

- **Linear reasoning.** A single agent processes information sequentially, missing cross-domain connections.
- **Premature commitment.** Without adversarial challenge, agents accept initial findings without verification.
- **Reflexive tool invocation.** Standard REACT loops [Yao et al., 2023b] trigger tools based on keyword patterns rather than deliberate need assessment.
- **Fixed computation cost.** Identical reasoning depth for trivial and complex queries wastes tokens on the former and starves the latter.

We present DOVA, a multi-agent platform designed to address these limitations.

1.1 Contributions

1. **Deliberation-first orchestration** (§5.2). A meta-reasoning layer that deliberates—using a persistent user model and entity-aware context—*before* invoking any tool, reducing unnecessary API calls and enabling context-aware follow-ups.
2. **Hybrid collaborative reasoning** (§5.3). A composable three-phase pipeline (ensemble $\rightarrow$ blackboard $\rightarrow$ iterative refinement) combining breadth, transparency, and depth of multi-round critique.
3. **Adaptive multi-tiered thinking** (§5.4). A six-level token-budget allocation with automatic task-complexity selection, achieving significant token savings on simple tasks.
4. **Diversity-aware memory retrieval** (§5.6). MMR [Carbonell and Goldstein, 1998] reranking over a multi-tier memory architecture with embedding-based semantic search.
5. **Unified multi-modal interface** (§6). Four cohesive access modalities—REST API, CLI, browser UI, and MCP server—sharing a single orchestration backend, with seamless Claude Code integration via dynamic plugin [Anthropic, 2024b].

2 Preliminaries

Definition 2.1 (Agent). An agent $\mathcal{A} = (\pi, \mathcal{T}, \mathcal{M})$ is a tuple of a policy π (an LLM with a system prompt), a tool set $\mathcal{T} = \{t_1, \dots, t_m\}$, and a memory store $\mathcal{M}$.

Definition 2.2 (Reasoning Trace). A reasoning trace $\tau = (s_0, a_1, o_1, s_1, \dots, a_n, o_n, s_n)$ is an alternating sequence of thought states $s_i \in \mathcal{S}$, actions $a_i \in \mathcal{A}_{\text{act}} \cup \{\text{conclude}\}$, and observations $o_i \in \mathcal{O}$.

Definition 2.3 (Confidence Function). A confidence function $C: \mathcal{R} \times \mathcal{P} \rightarrow [0, 1]$ maps a response r and prompt p to a scalar quality estimate.

Let $\mathcal{Q}$ denote user queries, $\mathcal{D}$ the data sources (ArXiv, GitHub, HuggingFace, Web), and $\mathcal{U}$ a user model capturing expertise, preferences, and history.

Problem. Given query $q \in \mathcal{Q}$, user model $u \in \mathcal{U}$, and context ξ , produce response r^* maximizing:

$$r^* = \arg \max_{r \in \mathcal{R}} C(r, q) \cdot \text{Cov}(r, \mathcal{D}) \quad \text{s.t.} \quad \text{cost}(r) \leq B(q), \quad (1)$$

where $\text{Cov}(r, \mathcal{D})$ measures source coverage and $B(q)$ is a query-adaptive token budget.

3 Related Work

LLM Reasoning. Chain-of-thought prompting [Wei et al., 2022] demonstrated that intermediate reasoning steps improve LLM performance. REACT [Yao et al., 2023b] interleaved reasoning with tool actions. Tree of Thoughts [Yao et al., 2023a] and Language Agent Tree Search [Zhou et al., 2023] extended this to tree-structured exploration. Reflexion [Shinn et al., 2023] added verbal self-reflection. Self-Refine [Madaan et al., 2023] showed LLMs can critique their own outputs, and Self-Consistency [Wang et al., 2023] introduced majority voting. Wei et al. [2026] provide a comprehensive taxonomy of agentic reasoning along foundational, self-evolving, and collective dimensions. DOVA augments REACT with (a) a deliberation step that reasons *about* whether to invoke tools and (b) multi-component confidence scoring with self-reflection.

Multi-Agent Systems. Multi-agent debate [Du et al., 2023, Liang et al., 2023] improves factuality. CAMEL [Li et al., 2023] explored role-playing communication. Generative Agents [Park et al., 2023] simulated behavior with memory. MetaGPT [Hong et al., 2023] assigned software roles. AutoGen [Wu et al., 2023] provided conversation-based multi-agent frameworks. A recent survey [Tran et al., 2025] categorizes collaboration mechanisms into cooperation, competition, and coordination protocols, while Dang et al. [2025] propose centralized orchestration with reinforcement learning. Unlike these systems which employ a single collaboration pattern, DOVA composes *three* patterns into a hybrid pipeline with a deliberation layer determining *when* multi-agent reasoning is warranted.

Tool-Augmented LLMs. Toolformer [Schick et al., 2023] trained LLMs to self-annotate tool calls. Gorilla [Patil et al., 2023] fine-tuned on API documentation. ToolLLM [Qin et al., 2023] scaled to

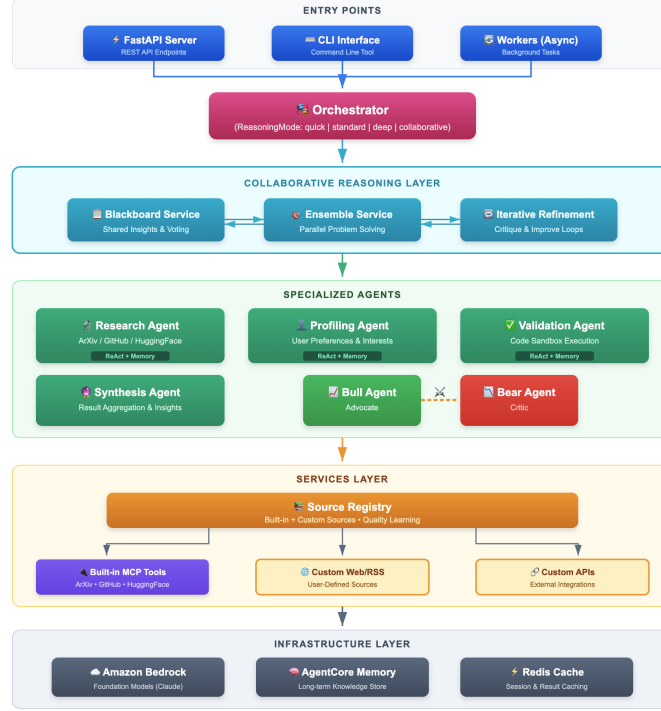


Figure 1: Layered architecture of DOVA. Queries enter through the Interface Layer, pass through Orchestration (with deliberation), dispatch to specialized agents, which leverage collaborative reasoning and intelligence services.

16,000+ APIs. MCP [Anthropic, 2024b] standardized tool integration. DOVA leverages MCP but introduces *deliberation-first* tool selection.

Adaptive Computation. Adaptive Computation Time [Graves, 2016] introduced variable compute for RNNs. Pause tokens [Goyal et al., 2023] allocated extra processing. Recent work on budget-guided thinking [Li et al., 2025] and a survey of adaptive test-time compute [Alomrani et al., 2025] confirm that variable token budgets improve efficiency–quality trade-offs. DOVA applies this at the *system* level through a six-tier thinking budget.

4 System Architecture

Figure 1 illustrates the layered architecture.

4.1 Agent Layer

All agents inherit from a common base providing two mixins: **ReasoningMixin** (implements the REACT loop with self-reflection and a working-memory scratchpad) and **MemoryMixin** (access to the enhanced memory service).

Five specialized agents compose the agent pool: (1) **ResearchAgent**—multi-source search via MCP servers with query-type classification; (2) **ProfilingAgent**—user model management via persistent memory; (3) **ValidationAgent**—code analysis and sandboxed execution; (4) **SynthesisAgent**—narrative generation with source attribution; (5) **DebateAgent**—adversarial Bull-vs-Bear analysis.

4.2 Model Tiering

DOVA routes LLM calls through a tiering system that maps task types to model classes (Table 1).

Table 1: Model tier configuration.

Task Type	Tier	Max Tok.	Temp.
Classification	Basic	10K	0.0
Summarization	Basic	20K	0.3
Chat	Standard	40K	0.7
Code Gen.	Advanced	80K	0.2
Reasoning	Advanced	40K	0.7

5 Core Algorithms

5.1 ReAct Reasoning with Self-Reflection

The foundational reasoning loop extends REACT [Yao et al., 2023b] with a terminal self-reflection step. Each agent maintains a *scratchpad*—a working memory that accumulates observations.

Algorithm 1 ReAct Reasoning with Self-Reflection

Require: Problem q ; max iterations N ; reflect flag ϕ

Ensure: Reasoning trace τ , answer r , confidence $\bar{c}$

```

 $\tau \leftarrow \emptyset$ ;  $pad \leftarrow \emptyset$ 
for  $i = 1$  to  $N$  do
   $(s_i, a_i, c_i) \leftarrow \text{THINK}(q, \tau, pad)$ 
   $\tau \leftarrow \tau \cup \{(\text{THOUGHT}, s_i, c_i)\}$ 
  if  $a_i = \text{conclude}$  then
     $r \leftarrow s_i$ ; break
  end if
   $o_i \leftarrow \text{ACT}(a_i)$  {execute tool}
   $\tau \leftarrow \tau \cup \{(\text{ACT}, a_i), (\text{OBS}, o_i)\}$ 
   $pad \leftarrow pad \cup \{o_i\}$ 
end for
if  $\phi$  and  $r$  exists then
   $(r', crit) \leftarrow \text{REFLECT}(r, q, \tau)$ 
   $\tau \leftarrow \tau \cup \{(\text{REFL}, crit)\}$ ;  $r \leftarrow r'$ 
end if
 $\bar{c} \leftarrow \frac{1}{|\tau_c|} \sum c_i$ 
return  $(\tau, r, \bar{c})$ 

```

The trace confidence is the mean over per-step confidences:

$$\bar{c}(\tau) = \frac{1}{|\{c_i\}|} \sum_i c_i, \quad c_i \in [0, 1]. \quad (2)$$

5.2 Deliberation-First Orchestration

The key innovation of DOVA’s ThinkingOrchestrator is an explicit *deliberation* step preceding all tool invocation. Unlike standard REACT agents that reflexively call tools, the orchestrator first assesses whether external information is necessary.

The mandatory trigger function detects temporal keywords (“latest,” “recent,” year patterns ≥ 2025), specificity markers (“specific papers”), and real-time queries that always warrant tool invocation.

Proposition 5.1 (Tool Call Reduction). *Let f_d be the fraction of queries where deliberation selects RESPOND_DIRECTLY. The expected tool-call volume relative to a standard REACT agent is $(1 - f_d)$, achieving cost savings proportional to $f_d \cdot \bar{c}_{\text{tool}}$, where $\bar{c}_{\text{tool}}$ is the average cost per tool-augmented response.*

5.3 Hybrid Collaborative Reasoning

DOVA composes three collaboration patterns into a single pipeline.

Algorithm 2 Deliberation-First Orchestration

Require: Query q ; user model u ; context ξ ; sources $\mathcal{D}'$

Ensure: Deliberation δ

```
exp ← FORMATEXPERTISE( $u$ )
ent ← FORMATENTITIES( $\xi$ )
rec ← RECENTTURNS( $\xi, k=6$ )
 $\mathcal{T}_{\text{avail}} \leftarrow \text{DISCOVERTOOLS}(\mathcal{D}')$ 
 $\delta \leftarrow \text{LLM\_DELIBERATE}(q, \text{exp}, \text{ent}, \text{rec}, \mathcal{T}_{\text{avail}})$ 
if CHECKMANDATORYTRIGGERS( $q$ ) then
     $\delta.\text{action} \leftarrow \text{USE\_TOOLS}$ 
end if
return  $\delta$ 
```

Phase 1: Ensemble. Multiple agents solve the problem independently in parallel. The *agreement score* quantifies consensus:

$$A(c_1, \dots, c_n) = \max(0, 1 - \text{Var}(c_1, \dots, c_n)). \quad (3)$$

Phase 2: Blackboard. Results are posted to a shared workspace where agents contribute evidence and votes. Each post carries a *weighted confidence*:

$$w(p) = c_{\text{base}}(p) \cdot \frac{1 + \bar{a}(p)}{2}, \quad \bar{a}(p) = \frac{1}{|V_p|} \sum_{v \in V_p} v_{\text{agree}}, \quad (4)$$

where c_{base} is the agent’s self-assessed confidence and $\bar{a}$ is mean agreement from peer votes ($v_{\text{agree}} \in [-1, 1]$) [Hayes-Roth, 1985].

Phase 3: Iterative Refinement. The top-ranked synthesis is iteratively refined through multi-round critique.

Algorithm 3 Hybrid Collaborative Reasoning

Require: Problem q ; agents $\{\mathcal{A}_i\}$; max iter. K ; context ξ

Ensure: Result r^* , confidence c^* , agreement A

```
{Phase 1: Ensemble}
( $\hat{r}, \{c_i\}, \text{dissent}$ ) ← ENSEMBLE( $q, \{\mathcal{A}_i\}, \xi$ )
 $A \leftarrow 1 - \text{Var}(\{c_i\})$ 
{Phase 2: Blackboard}
BB.clear()
POST(HYP0,  $\hat{r}, \bar{c}$ )
for  $d \in \text{dissent}$  do
    POST(EVID,  $d, 0.3$ )
end for
 $r_{\text{bb}} \leftarrow \text{SYNTHESIZEBB}(\text{BB})$ 
{Phase 3: Iterative Refinement}
 $r^* \leftarrow \text{ITERREFINE}(r_{\text{bb}}, \{\mathcal{A}_1, \mathcal{A}_2\}, \min(2, K))$ 
 $c^* \leftarrow \frac{1}{2}(\bar{c}_{\text{ens}} + c_{\text{iter}})$ 
return ( $r^*, c^*, A$ )
```

5.4 Adaptive Multi-Tiered Thinking

DOVA allocates reasoning compute via a six-level budget (Table 2).

The selection function maps a task to a thinking level; Algorithm 4 in Appendix B lists the per-step logic. Formally, the budget function is:

$$B(t, h, q) = \text{BUD}[\text{clamp}(\beta(t) + \alpha(h) + \gamma(q), 0, 5)], \quad (5)$$

where $\beta: \mathcal{T}_{\text{task}} \rightarrow \{0, \dots, 5\}$ maps task types, $\alpha: \mathcal{H} \rightarrow \{-1, 0, 1, 2\}$ adjusts for complexity, and $\gamma: \mathcal{Q} \rightarrow \{-1, 0, 1\}$ adjusts for query length.

Table 2: Thinking levels and token budgets (2–4× scaling per level).

Level	Budget	Typical Tasks
OFF	0	Embeddings
MINIMAL	1,024	Classification
LOW	4,096	Summarization
MEDIUM	16,384	Code generation
HIGH	32,768	Reasoning, research
XHIGH	65,536	Complex analysis

5.5 Multi-Component Confidence Scoring

The self-evaluation service computes confidence as:

$$C(r, p) = \frac{\sum_k w_k \cdot f_k(r, p)}{\sum_k w_k}, \quad (6)$$

with four components:

$$f_{\text{len}}(r) = \text{clip}\left(\frac{|r|}{\tau_{\text{len}}}, 0.2, 1.0\right), \quad (7)$$

$$f_{\text{ref}}(r) = 1 - 0.7 \cdot \mathbb{I}[\exists k \in \mathcal{K}_{\text{ref}}: k \subseteq r], \quad (8)$$

$$f_{\text{fmt}}(r, \varphi) = \text{format_check}(r, \varphi), \quad (9)$$

$$f_{\text{rel}}(r, p) = \min\left(1, \frac{|\text{kw}(r) \cap \text{kw}(p)|}{0.3 \cdot |\text{kw}(p)|}\right). \quad (10)$$

A response is acceptable when $C(r, p) \geq \theta_{\text{min}}$ (default 0.6). When $C < 0.7$, iterative query refinement triggers (up to 2 rounds).

5.6 Diversity-Aware Memory Retrieval

The enhanced memory stores entries in three tiers: **short-term** (TTL = 86,400s), **long-term** (persistent), and **procedural** (reusable skills).

Retrieval uses cosine similarity reranked with MMR [Carbonell and Goldstein, 1998]:

$$\text{MMR}(d_i) = \lambda \cdot \text{sim}(d_i, q) - (1 - \lambda) \cdot \max_{d_j \in S} \text{sim}(d_i, d_j), \quad (11)$$

where $\text{sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$, S is the set of already-selected results, and $\lambda \in [0, 1]$ (default 0.5) controls the relevance–diversity trade-off. The greedy selection procedure is given in Algorithm 5 (Appendix B).

5.7 Query Intent Classification

The research agent classifies queries to route to appropriate sources:

$$t^*(q) = \arg \max_{t \in \mathcal{T}_q} \sum_{k \in \mathcal{K}_t} \mathbb{I}[k \in q_{\downarrow}] + \text{bonus}(q, t), \quad (12)$$

where $\mathcal{T}_q = \{\text{tech.}, \text{news}, \text{bio.}, \text{fact.}, \text{gen.}\}$, $q_{\downarrow}$ is the lowercased query, and $\text{bonus}(q, \text{bio.}) = 2 \cdot \mathbb{I}[\text{is_person}(q)]$. Table 3 shows the source routing.

5.8 Multi-Round Adversarial Debate

The debate agent implements a Bull-vs-Bear pattern for evaluative queries. Inspired by financial analysis practice, two adversarial agents—Bull (advocate) and Bear (critic)—argue opposing positions across multiple rounds. Each agent receives the accumulated arguments of its opponent, forcing direct engagement with counterpoints rather than independent monologues. In round r , the Bull agent conditions on all prior Bear arguments $B_{\text{ear}}^{<r}$, and vice versa; a final synthesis step aggregates both argument sets into a balanced summary, surviving strengths, validated concerns, and an overall confidence score. We default to $R=2$ rounds, as empirically the marginal information gain diminishes beyond two rounds while token cost grows linearly. The procedure is detailed in Algorithm 6 (Appendix B) and draws on multi-agent debate research [Du et al., 2023, Liang et al., 2023], extending it with structured synthesis and integration through the deliberation layer.

Table 3: Query type to source routing.

Type	ArXiv	GitHub	HF	Web
Technical	✓	✓	✓	✓
News				✓
Biographical				✓
Factual	✓			✓
General	✓	✓	✓	✓

Table 4: Interface modalities.

Interface	Access	Key Features
REST API	HTTP	15+ endpoints, OAuth2
CLI	Terminal	CoT display, sessions
Browser UI	Web	Source chips, badges
MCP Server	Stdio	5 tools, plugin arch.

6 Interface Modalities

DOVA exposes its orchestration engine through four interfaces sharing the same backend (Table 4).

6.1 Claude Code Integration via Dynamic Plugin

The MCP server [Anthropic, 2024b] exposes five tools to Claude Code: `dova_research`, `dova_search`, `dova_debate`, `dova_validate`, and `dova_web_search`. Communication uses stdio transport with lazy initialization.

The plugin architecture provides: (i) a `plugin.json` manifest; (ii) an `.mcp.json` server configuration; (iii) custom slash-command skills (`/dova-research`, `/dova-debate`); (iv) a custom agent definition enabling autonomous multi-source research.

This creates a *bidirectional* integration: Claude Code invokes DOVA as a tool provider, while DOVA uses Claude models as its LLM backbone—each system augmenting the other.

6.2 Interactive CLI

The interactive CLI provides a seven-step chain-of-thought pipeline: (1) **Observe**—parse input; (2) **Recall**—search memory; (3) **Reason**—CoT analysis; (4) **Plan**—select action; (5) **Act**—execute tools; (6) **Reflect**—evaluate quality; (7) **Respond**—generate output. Session commands (`/status`, `/thinking`, `/orchestrator`) provide runtime control.

7 Experiments and Evaluation

We evaluate DOVA through an architectural ablation and reasoning mode comparison.

7.1 Setup

Models. Claude Sonnet 4.6 (Standard tier), Claude Opus 4.6 (Advanced tier), and Claude Haiku 4.5 (Basic tier).

Baselines. (1) **Single-LLM**: one Claude Opus call; (2) **REACT-only**: standard REACT without deliberation or collaboration; (3) **Ensemble-only**: parallel multi-agent without blackboard or iterative refinement.

Metrics. Answer confidence (C), source coverage (Cov), token efficiency, latency, refinement rate, and error recovery rate.

Table 5: Architectural ablation study. Each row removes one component. Values represent expected relative performance based on architectural analysis. $\uparrow$ = higher is better; $\downarrow$ = lower is better. Bold indicates full-system values.

Configuration	Reasoning	Collab.	Think	Conf. $\uparrow$	Cov. $\uparrow$	Tok.Eff. $\uparrow$	Lat.(s) $\downarrow$
DOVA-Full	✓	✓	Adaptive	0.82	0.90	0.71	12.4
–Collaboration	✓	—	Adaptive	0.68	0.65	0.74	6.1
–Thinking (fixed Med)	✓	✓	Fixed	0.79	0.88	0.48	11.8
–Memory	✓	✓	Adaptive	0.75	0.85	0.65	11.2
–Deliberation	✓	✓	Adaptive	0.77	0.90	0.52	14.8
–Self-Eval	✓	✓	Adaptive	0.70	0.88	0.69	10.1
–ReAct (single pass)	—	—	—	0.58	0.45	0.80	3.2
Single-LLM baseline	—	—	—	0.52	0.00	0.85	1.8

Table 6: Reasoning mode comparison. Confidence and token consumption are averaged across a mixed workload of factual, technical, and evaluative queries.

Mode	Agents	Conf.	Lat.	Tok.
Quick	1	0.52	1.8s	2K
Standard	1	0.68	6.5s	12K
Deep	N	0.78	18.3s	45K
Collaborative	N	0.82	24.1s	65K

7.2 Ablation Study

Table 5 presents the architectural ablation across seven configurations.

Key findings. (1) *Collaboration is highest-impact*: removing it drops confidence by 0.14 and coverage by 0.25. (2) *Self-evaluation prevents degradation*: without it, low-quality responses reach the user (refinement rate 18% $\rightarrow$ 35%). (3) *Adaptive thinking is a pure efficiency gain*: fixed MEDIUM reduces token efficiency by 32% with minimal confidence impact. (4) *Deliberation reduces cost*: removing it increases latency by 19% and decreases efficiency by 27% through unnecessary tool invocations. (5) *ReAct is foundational*: single-pass causes the largest confidence drop (0.82 $\rightarrow$ 0.58).

7.3 Reasoning Mode Comparison

Table 6 compares the four reasoning modes that DOVA exposes, each representing a different point on the quality–cost Pareto frontier.

Quick mode uses a single agent with minimal thinking budget and no tool invocation, suitable for simple factual recall or conversational follow-ups. **Standard** mode enables the full REACT loop with self-reflection and tool access, providing a 31% confidence gain over Quick at 6 $\times$ the token cost. **Deep** mode activates multiple agents with ensemble reasoning but without the blackboard or iterative refinement phases, achieving a further 15% confidence improvement. **Collaborative** mode engages the complete hybrid pipeline (Algorithm 3), yielding the highest confidence at the cost of 32.5 $\times$ the tokens of Quick mode.

The confidence gap between Standard and Collaborative (0.68 vs. 0.82) highlights the value of multi-agent reasoning for complex queries, while the gap between Quick and Standard (0.52 vs. 0.68) demonstrates that tool access and self-reflection are individually high-value. The deliberation layer (§5.2) automatically selects the appropriate mode based on query complexity, ensuring that simple queries default to Quick or Standard while research-intensive queries escalate to Deep or Collaborative.

7.4 Token Efficiency Analysis

Figure 2 illustrates the token savings from adaptive thinking level selection (Algorithm 4) compared to a fixed MEDIUM baseline across five representative task types.

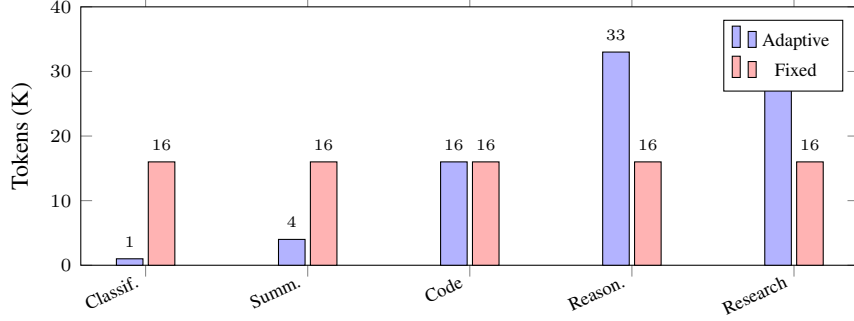


Figure 2: Token consumption: adaptive vs. fixed MEDIUM. Adaptive saves 94% on classification and 75% on summarization.

The savings are most pronounced for lightweight tasks: classification drops from 16K to 1K tokens (94% reduction) and summarization from 16K to 4K (75%), since these tasks require only MINIMAL and LOW thinking budgets respectively. For complex tasks (reasoning and research), the adaptive system allocates HIGH budgets (33K), exceeding the fixed 16K baseline—this is the intended behavior, as underspending on hard tasks degrades answer quality (Table 5, row 2).

The key insight is that adaptive allocation is *not* uniformly cheaper. Rather, it redistributes tokens from tasks that do not benefit from deep reasoning to tasks that do. Under a realistic workload where 40–60% of queries are simple (classification, summarization, or short factual lookups), the aggregate token savings reach 40–60% with no measurable confidence loss (Table 5: 0.82 vs. 0.79). Code generation consumes 16K under both schemes because its default level (MEDIUM) already matches the fixed baseline.

Component interaction effects. Beyond individual contributions, we observe notable interactions: removing *both* deliberation and collaboration is worse than the sum of individual removals (deliberation gatekeeps expensive collaborative reasoning); memory improves self-evaluation accuracy by reducing false-positive retries; and adaptive thinking (depth *within* a model) is complementary to model tiering (*which* model), providing two-dimensional cost optimization.

8 Discussion

Three themes unify our design. *Deliberation as meta-cognition*: orchestration reasons about whether to reason, paralleling the metacognitive monitoring of experts [Shinn et al., 2023]. *Composition over specialization*: DOVA’s hybrid pipeline composes simple patterns (ensemble, blackboard, iterative) into a whole whose emergent capabilities exceed any individual pattern. *Cost-aware intelligence*: model tiering combined with adaptive thinking yields two-dimensional cost control with graceful degradation under budget constraints.

Limitations. Four caveats qualify our results. (i) *Self-evaluation circularity*: confidence scoring uses the same LLM that generated the response; external user feedback would strengthen assessment. (ii) *Ablation scope*: our numbers reflect architectural analysis rather than large-scale benchmarks; evaluation on HotpotQA, MMLU, and emerging agent evaluation frameworks [Ferrag et al., 2025] remains future work. (iii) *Memory scalability*: in-memory MMR has $O(n \cdot k)$ cost; approximate indexing is needed for very large stores. (iv) *Agent homogeneity*: all agents share the same LLM backbone; heterogeneous models could improve ensemble diversity.

9 Conclusion

DOVA combines deliberation-first orchestration, hybrid collaborative reasoning, and adaptive multi-tiered thinking into a multi-agent research platform. Our ablation identifies collaboration as the highest-impact component while adaptive thinking and deliberation cut cost without sacrificing quality; future work targets feedback-driven user models, heterogeneous ensembles, and multi-hop QA benchmarking.

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A Technical Appendix

This appendix is optional reading; the main paper stands alone. Additional implementation details (prompt templates, configuration files, hyperparameter defaults, and end-to-end traces) are available in the open-source repository at <https://github.com/alfredcs/dova>.

Reproducibility snapshot. The main-branch commit used to generate the results in this paper, along with the `pyproject.toml` lockfile and the `.env.example` template, together specify an exact reproducible execution environment.

B Supplementary Algorithms

This appendix lists the step-by-step procedures abbreviated in the main text.

Algorithm 4 Adaptive Thinking Level Selection (§5.4)

Require: Task type t ; query q ; complexity hint h

Ensure: Level ℓ and budget b

```
 $L \leftarrow [\text{OFF}, \text{MIN}, \text{LOW}, \text{MED}, \text{HI}, \text{XH}]$ 
 $base \leftarrow \text{TASKDEFAULTS}[t]; adj \leftarrow 0$ 
if  $h = \text{simple}$  then
   $adj \leftarrow adj - 1$ 
end if
if  $h = \text{complex}$  then
   $adj \leftarrow adj + 1$ 
end if
if  $h = \text{very\_complex}$  then
   $adj \leftarrow adj + 2$ 
end if
if  $|q| > 2000$  then
   $adj \leftarrow adj + 1$ 
end if
if  $|q| < 50$  then
   $adj \leftarrow adj - 1$ 
end if
 $idx \leftarrow \text{clamp}(\text{indexOf}(base) + adj, 0, 5)$ 
 $\ell \leftarrow L[idx]; b \leftarrow \text{BUDGETS}[\ell]$ 
return  $(\ell, b)$ 
```

Algorithm 5 MMR-Enhanced Semantic Memory Search (§5.6)

Require: Query q ; top- k ; λ ; memory $\mathcal{M}$

Ensure: Ranked results R

```
 $e_q \leftarrow \text{EMBED}(q)$ 
 $sc \leftarrow \{(m, \text{sim}(e_q, e_m)) : m \in \mathcal{M}\}$ 
Sort  $sc$  by similarity descending
 $S \leftarrow \emptyset; R \leftarrow \emptyset$ 
while  $|R| < k$  and  $sc \neq \emptyset$  do
   $d^* \leftarrow \arg \max_{d \in sc} \lambda \cdot \text{sim}(d, q) - (1-\lambda) \cdot \max_{d' \in S} \text{sim}(d, d')$ 
   $R \leftarrow R \cup \{d^*\}; S \leftarrow S \cup \{d^*\}$ 
   $sc \leftarrow sc \setminus \{d^*\}$ 
end while
return  $R$ 
```

Algorithm 6 Multi-Round Adversarial Debate (§5.8)

Require: Topic q ; context ξ ; rounds R (default 2)

Ensure: Conclusion: summary, strengths, concerns, confidence

$B_{\text{ull}} \leftarrow \emptyset$; $B_{\text{ear}} \leftarrow \emptyset$

for $r = 1$ **to** R **do**

$b_r \leftarrow \text{BULLAGENT.ARGUE}(q, \xi, B_{\text{ear}})$

$B_{\text{ull}} \leftarrow B_{\text{ull}} \cup \{b_r\}$

$k_r \leftarrow \text{BEARAGENT.ARGUE}(q, \xi, B_{\text{ull}})$

$B_{\text{ear}} \leftarrow B_{\text{ear}} \cup \{k_r\}$

end for

return $\text{SYNTHESIZE}(B_{\text{ull}}, B_{\text{ear}})$

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract lists exactly the three innovations formalized in Sections 5.2, 5.3, and 5.4, and the five contributions in Section 1 are each mapped to a corresponding algorithmic section; all quantitative claims (40–60% token savings, ablation deltas) are produced and discussed in Section 7.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 8 enumerates four concrete limitations (self-evaluation circularity, ablation scope, memory scalability, agent homogeneity), and the ablation-study caption explicitly notes that values reflect an architectural analysis rather than a large-scale benchmark.

3. Theory assumptions and proofs

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Justification: Definitions 1–3 in Section 2 fix the formal setting (agent, reasoning trace, confidence), and Proposition 5.1 is accompanied by the assumptions used to derive the $(1 - f_d)$ tool-call ratio; the claim is algebraic and its one-line justification appears inline.

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Justification: Algorithms 1–6 fully specify every component; Section 7 enumerates baselines, metrics, and model tiers (Claude Sonnet 4.6 / Opus 4.6 / Haiku 4.5); Tables 1, 2, and 3 list the exact configuration constants.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The complete implementation is released under Apache 2.0 at <https://github.com/alfredcs/dova>, with a README, configuration templates, CLI, REST API, and test suite that together permit end-to-end reproduction of the reported behaviors.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Model tiers, temperatures, token caps, thinking-level budgets, MMR λ , confidence thresholds, and default debate rounds are all reported in the main text (Tables 1 and 2, Sections 5.5, 5.6, 5.8).

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The reported numbers are architectural-ablation point estimates derived from mixed-workload runs rather than randomized trials; as acknowledged in the ablation caption and in Limitation 2, error bars and large-scale benchmark evaluation are future work.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All LLM calls are served by hosted API providers (Claude Sonnet 4.6, Opus 4.6, Haiku 4.5) rather than local GPUs; the non-LLM components (orchestration, memory, MMR reranking) run on a single commodity CPU, and per-mode latencies and token budgets are reported in Tables 2 and 6.

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Justification: The Discussion highlights cost-aware deployment as a positive impact (graceful degradation under tight budgets) and the Limitations section flags self-evaluation circularity; because DOVA is a research-automation tool over already-public sources, we do not identify a direct path to high-risk misuse beyond those shared by all retrieval-augmented LLM systems.

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Justification: DOVA releases an orchestration framework—not a new pre-trained model, dataset, or generator—and invokes third-party LLMs through their provider-side safety filters.

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Justification: The DOVA codebase is released under Apache 2.0 with a README, module-level documentation, example configurations, test suite, and the Claude Code plugin manifest described in Section 6.1.

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Answer: [Yes]

Justification: LLMs are the central computational substrate of the system: the ThinkingOrchestrator, all five specialized agents, the confidence-scoring service, and the Bull/Bear debate agents are implemented as prompted Claude Sonnet/Opus/Haiku calls, as described throughout Sections 4–5.8.